

Analysis of Exact and ROM Bode Responses for Alternative State Vector Definitions

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Summary

More than one state vector definition can typically be used to describe any given dynamical system. The ASWING dynamical system is one such example, in which the global frame accelerations can be either included or left out of the state vector.

In both the time or frequency domain, the alternative state vector definitions, if correct and complete, will produce the same output results. However, when an eigenmode-based Reduced Order Model (ROM) is used to simplify the system, for some state vector choices there can be some ambiguity in how some of the BODE response quantities are computed. Furthermore, the different options for computing some of the responses will give different results. In the ASWING system, the acceleration and rigid-body displacement responses have such ambiguities.

The objective of this document is to examine these ambiguities via a simple model system which has closed-form solutions. Comparison of the exact responses and the ROM responses will indicate which of the alternative ROM response calculation methods should be used for best ROM accuracy in ASWING.

1 General Dynamical System

1.1 Direct analysis

In the general state space approach, the evolution equations and output equations are assumed to have the form

$$\begin{aligned} f(\mathbf{x}, \dot{\mathbf{x}}, \mathbf{u}) &= 0 \\ \mathbf{y} &= g(\mathbf{x}, \dot{\mathbf{x}}, \mathbf{u}) \end{aligned}$$

where $\mathbf{x}$ is the state vector, $\mathbf{u}$ is the control vector, and $\mathbf{y}$ is the output vector. In linearized form, these are

$$\begin{aligned} \mathbf{M} \dot{\mathbf{x}} &= \mathbf{A} \mathbf{x} + \mathbf{B} \mathbf{u} \\ \mathbf{y} &= \mathbf{C} \mathbf{x} + \mathbf{Q} \dot{\mathbf{x}} + \mathbf{D} \mathbf{u} \end{aligned}$$

The additional $\mathbf{Q} \dot{\mathbf{x}}$ term in the output equation, which normally does not appear in traditional state-space analyses, is included here to allow for a system whose mass matrix $\mathbf{M}$ does not have full rank. In this case $\dot{\mathbf{x}}$ cannot be explicitly given in terms of $\mathbf{x}$ and $\mathbf{u}$. The ASWING dynamical system is one example of this situation.

The Bode response $\mathbf{x} = \hat{\mathbf{x}} \exp(i\omega t)$, $\mathbf{y} = \hat{\mathbf{y}} \exp(i\omega t)$ to some forcing $\mathbf{u} = \hat{\mathbf{u}} \exp(i\omega t)$ is computed by solving the following linear system for $\hat{\mathbf{x}}$, and then using this result to obtain $\hat{\mathbf{y}}$.

$$i\omega \mathbf{M} \hat{\mathbf{x}} = \mathbf{A} \hat{\mathbf{x}} + \mathbf{B} \hat{\mathbf{u}} \tag{1}$$

$$\hat{\mathbf{x}} = [i\omega \mathbf{M} - \mathbf{A}]^{-1} \mathbf{B} \hat{\mathbf{u}} \tag{1}$$

$$\hat{\mathbf{y}} = [\mathbf{C} + i\omega \mathbf{Q}] \hat{\mathbf{x}} + \mathbf{D} \hat{\mathbf{u}} \tag{2}$$

1.2 Eigenmode ROM analysis

The system eigenvalue matrix Λ , and left and right eigenvector matrices $\mathbf{V}$ and $\mathbf{W}$ satisfy

$$\begin{aligned}\mathbf{M}\mathbf{V}\Lambda &= \mathbf{A}\mathbf{V} \\ \mathbf{W}^*\mathbf{M}\Lambda &= \mathbf{W}^*\mathbf{A} \\ \mathbf{W}^*\mathbf{M}\mathbf{V} &= \mathbf{I} \\ \mathbf{W}^*\mathbf{A}\mathbf{V} &= \Lambda\end{aligned}$$

An eigenmode expansion for the state is

$$\mathbf{x} = \mathbf{V}\mathbf{z}$$

which reduces the evolution and output equations to the following.

$$\begin{aligned}\dot{\mathbf{z}} &= \Lambda\mathbf{z} + \mathbf{B}'\mathbf{u} \\ \mathbf{y} &= \mathbf{C}'\mathbf{z} + \mathbf{D}'\mathbf{u} \\ \mathbf{B}' &= \mathbf{W}^*\mathbf{B} \\ \mathbf{C}' &= \mathbf{C}\mathbf{V} + \mathbf{Q}\mathbf{V}\Lambda \\ \mathbf{D}' &= \mathbf{D} + \mathbf{Q}\mathbf{V}\mathbf{W}^*\mathbf{B}\end{aligned}$$

The Bode response computed using the ROM basis is as follows.

$$\begin{aligned}i\omega\hat{\mathbf{z}} &= \Lambda\hat{\mathbf{z}} + \mathbf{W}^*\mathbf{B}\hat{\mathbf{u}} \\ \hat{\mathbf{z}} &= [i\omega\mathbf{I} - \Lambda]^{-1}\mathbf{W}^*\mathbf{B}\hat{\mathbf{u}}\end{aligned}\tag{3}$$

$$\hat{\mathbf{x}} = \mathbf{V}\hat{\mathbf{z}}\tag{4}$$

$$\hat{\mathbf{y}} = \mathbf{C}'\hat{\mathbf{z}} + \mathbf{D}'\hat{\mathbf{u}}\tag{5}$$

If the chosen eigenmode basis $\mathbf{V}$ has full rank, the final Bode results (4),(5) will match the exact results (1),(2). The more interesting case is when $\mathbf{V}$ does not have full rank, which of course is the objective of a ROM in the first place. This case will be examined next.

2 Simple Example - I

2.1 Direct analysis

We now examine the time-evolution system used for time-marching OPER, BODE, and MODE, for the simplest possible case. The aircraft is idealized as a particle with position R , velocity U , and acceleration a . The force/mass F depends only on the velocity U and a control variable δ . We first use the standard dynamical state-space approach, with only R and U present in the state vector. The desired output is the acceleration.

$$\begin{aligned}\mathbf{x} &= \{R \ U\}^T \\ \mathbf{u} &= \{\delta\}^T \\ \mathbf{y} &= \{\dot{U}\}^T\end{aligned}$$

We assume simple dynamic equations of the form

$$\begin{aligned}\dot{R} &= U \\ \dot{U} &= F(U, \delta)\end{aligned}$$

where F is the force per mass. The system matrices then follow.

$$\begin{aligned}\mathbf{M} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} & \mathbf{A} &= \begin{bmatrix} 0 & 1 \\ 0 & F_U \end{bmatrix} & \mathbf{B} &= \begin{bmatrix} 0 \\ F_\delta \end{bmatrix} \\ \mathbf{C} &= \begin{bmatrix} 0 & 0 \end{bmatrix} & \mathbf{Q} &= \begin{bmatrix} 0 & 1 \end{bmatrix} & \mathbf{D} &= \begin{bmatrix} 0 \end{bmatrix}\end{aligned}$$

For computing the exact response to $\hat{\mathbf{u}} = \{\hat{\delta}\}^T$, the Bode system matrices and vectors are

$$\begin{aligned}[i\omega\mathbf{M} - \mathbf{A}] &= \begin{bmatrix} i\omega & -1 \\ 0 & i\omega - F_U \end{bmatrix} \\ \mathbf{B}\hat{\mathbf{u}} &= \begin{bmatrix} 0 \\ F_\delta \hat{\delta} \end{bmatrix} \\ [\mathbf{C} + i\omega\mathbf{Q}] &= \begin{bmatrix} 0 & i\omega \end{bmatrix}\end{aligned}$$

which are used in (1) to determine $\mathbf{x}$, and subsequently in (2) to determine $\mathbf{y}$.

$$\hat{\mathbf{x}} = \begin{bmatrix} \frac{1}{i\omega(i\omega - F_U)} \\ \frac{1}{i\omega - F_U} \end{bmatrix} F_\delta \hat{\delta} \quad (6)$$

$$\hat{\mathbf{y}} = \begin{bmatrix} \frac{i\omega}{i\omega - F_U} \end{bmatrix} F_\delta \hat{\delta} \quad (7)$$

These will be used as the reference solution for evaluating the best-possible accuracy of associated eigenmode Reduced Order Model (ROM) formulations.

2.2 ROM analysis

The eigenvalues and eigenvectors for this system are

$$\mathbf{\Lambda} = \begin{bmatrix} 0 & \\ & F_U \end{bmatrix} \quad \mathbf{V} = \begin{bmatrix} 1 & 1 \\ 0 & F_U \end{bmatrix} \quad \mathbf{W}^* = \begin{bmatrix} 1 & \frac{-1}{F_U} \\ 0 & \frac{1}{F_U} \end{bmatrix}$$

giving the following ROM matrices.

$$\begin{aligned} \mathbf{B}' &= \begin{bmatrix} \frac{-F_\delta}{F_U} \\ \frac{F_\delta}{F_U} \end{bmatrix} \\ \mathbf{C}' &= \begin{bmatrix} 0 & F_U^2 \end{bmatrix} \\ \mathbf{D}' &= \begin{bmatrix} F_\delta \end{bmatrix} \end{aligned}$$

The Bode response using the ROM is now computed via (3), (4), (5).

$$\begin{aligned} [i\omega\mathbf{I} - \mathbf{\Lambda}]^{-1} &= \begin{bmatrix} \frac{1}{i\omega} & 0 \\ 0 & \frac{1}{i\omega - F_U} \end{bmatrix} \\ \hat{\mathbf{z}} &= [i\omega\mathbf{I} - \mathbf{\Lambda}]^{-1} \mathbf{W}^* \mathbf{B} \hat{\mathbf{u}} = \begin{bmatrix} \frac{-1}{i\omega F_U} \\ \frac{1}{(i\omega - F_U)F_U} \end{bmatrix} F_\delta \hat{\delta} \\ \hat{\mathbf{x}} &= \mathbf{V} \hat{\mathbf{z}} = \begin{bmatrix} \frac{1}{i\omega(i\omega - F_U)} \\ \frac{1}{i\omega - F_U} \end{bmatrix} F_\delta \hat{\delta} \end{aligned} \tag{8}$$

$$\hat{\mathbf{y}} = \mathbf{C}' \hat{\mathbf{z}} + \mathbf{D}' \hat{\mathbf{u}} = \left[\frac{i\omega}{i\omega - F_U} \right] F_\delta \hat{\delta} \tag{9}$$

This ROM-based result (8),(9) exactly matches the direct result (6),(7). This is not surprising, since the eigenmode basis used here has full rank, and hence can represent the system perfectly with no approximation.

3 Simple Example - II

3.1 Direct analysis

We now examine the time-evolution of the same system, except that now the state vector is augmented to include the acceleration. The additional equation is simply the definition of this acceleration. Two different but formally equivalent output equations will also be considered.

$$\begin{aligned}\mathbf{x} &= \{R \ U \ a\}^T \\ \mathbf{u} &= \{\delta\}^T \\ \mathbf{y}_1 &= \{\dot{U}\}^T \\ \mathbf{y}_2 &= \{a\}^T\end{aligned}$$

The augmented dynamic equations are now

$$\begin{aligned}\dot{R} &= U \\ \dot{U} &= a \\ 0 &= F(U, \delta) - a\end{aligned}$$

giving the following system matrices.

$$\begin{aligned}\mathbf{M} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} & \mathbf{A} &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & F_U & -1 \end{bmatrix} & \mathbf{B} &= \begin{bmatrix} 0 \\ 0 \\ F_\delta \end{bmatrix} \\ \mathbf{C}_1 &= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} & \mathbf{Q}_1 &= \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} & \mathbf{D}_1 &= \begin{bmatrix} 0 \end{bmatrix} \\ \mathbf{C}_2 &= \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} & \mathbf{Q}_2 &= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} & \mathbf{D}_2 &= \begin{bmatrix} 0 \end{bmatrix}\end{aligned}$$

For computing the exact response to $\hat{\mathbf{u}} = \{\hat{\delta}\}^T$, the Bode system matrices and vectors are

$$\begin{aligned}[i\omega\mathbf{M} - \mathbf{A}] &= \begin{bmatrix} i\omega & -1 & 0 \\ 0 & i\omega & -1 \\ 0 & -F_U & 1 \end{bmatrix} \\ \mathbf{B} \hat{\mathbf{u}} &= \begin{bmatrix} 0 \\ 0 \\ F_\delta \hat{\delta} \end{bmatrix} \\ [\mathbf{C}_1 + i\omega\mathbf{Q}_1] &= \begin{bmatrix} 0 & i\omega & 0 \end{bmatrix} \\ [\mathbf{C}_2 + i\omega\mathbf{Q}_2] &= \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}\end{aligned}$$

which are used to determine $\mathbf{x}$ via (1), and subsequently $\mathbf{y}_1$ and $\mathbf{y}_2$ via (2).

$$\hat{\mathbf{x}} = \begin{bmatrix} \frac{1}{i\omega(i\omega - F_U)} \\ \frac{1}{i\omega - F_U} \\ \frac{i\omega}{i\omega - F_U} \end{bmatrix} F_\delta \hat{\delta} \quad (10)$$

$$\hat{\mathbf{y}}_1 = \hat{\mathbf{y}}_2 = \begin{bmatrix} \frac{i\omega}{i\omega - F_U} \end{bmatrix} F_\delta \hat{\delta} \quad (11)$$

As expected, these exactly match the previous analysis version results (6),(7).

3.2 ROM analysis

This system has the following eigenvalues and eigenvectors.

$$\mathbf{\Lambda} = \begin{bmatrix} 0 & \\ & F_U \end{bmatrix} \quad \mathbf{V} = \begin{bmatrix} 1 & \frac{1}{F_U} \\ 0 & 1 \\ 0 & F_U \end{bmatrix} \quad \mathbf{W}^* = \begin{bmatrix} 1 & 0 & \frac{-1}{F_U} \\ 0 & 1 & 1 \end{bmatrix}$$

Because of the singular mass matrix $\mathbf{M}$, the eigensystem is defective, with only two eigenvectors being available (same number as the rank of $\mathbf{M}$). The ROM matrices are computed as usual.

$$\begin{aligned} \mathbf{B}' &= \begin{bmatrix} \frac{-F_\delta}{F_U} \\ F_\delta \end{bmatrix} \\ \mathbf{C}'_1 = \mathbf{C}'_2 &= \begin{bmatrix} 0 & F_U \end{bmatrix} \\ \mathbf{D}'_1 &= \begin{bmatrix} F_\delta \end{bmatrix} \\ \mathbf{D}'_2 &= \begin{bmatrix} 0 \end{bmatrix} \end{aligned}$$

The Bode response using the ROM is computed as follows.

$$\begin{aligned} [i\omega\mathbf{I} - \mathbf{\Lambda}]^{-1} &= \begin{bmatrix} \frac{1}{i\omega} & 0 \\ 0 & \frac{1}{i\omega - F_U} \end{bmatrix} \\ \hat{\mathbf{z}} &= [i\omega\mathbf{I} - \mathbf{\Lambda}]^{-1} \mathbf{W}^* \mathbf{B} \hat{\mathbf{u}} = \begin{bmatrix} \frac{-1}{i\omega F_U} \\ \frac{1}{i\omega - F_U} \end{bmatrix} F_\delta \hat{\delta} \\ \hat{\mathbf{x}} &= \mathbf{V} \hat{\mathbf{z}} = \begin{bmatrix} \frac{1}{i\omega(i\omega - F_U)} \\ \frac{1}{i\omega - F_U} \\ \frac{F_U}{i\omega - F_U} \end{bmatrix} F_\delta \hat{\delta} \end{aligned} \tag{12}$$

$$\hat{\mathbf{y}}_1 = \mathbf{C}'_1 \hat{\mathbf{z}} + \mathbf{D}'_1 \hat{\mathbf{u}} = \begin{bmatrix} \frac{i\omega}{i\omega - F_U} \end{bmatrix} F_\delta \hat{\delta} \tag{13}$$

$$\hat{\mathbf{y}}_2 = \mathbf{C}'_2 \hat{\mathbf{z}} + \mathbf{D}'_2 \hat{\mathbf{u}} = \begin{bmatrix} \frac{F_U}{i\omega - F_U} \end{bmatrix} F_\delta \hat{\delta} \tag{14}$$

This ROM-based $\hat{\mathbf{x}}$ response does not match the exact direct result (10) in the third acceleration component. The second output form $\hat{\mathbf{y}}_2 = \hat{a}$ in (14) is therefore also incorrect. However, the first output form $\hat{\mathbf{y}}_1 = i\omega\hat{U}$ in (13) is correct. The following conclusions can be drawn for a system which contains the acceleration in the state vector, and hence has a singular mass matrix $\mathbf{M}$:

1. The ROM-approximated Bode-response state vector $\hat{\mathbf{x}}$ will not have an accurate acceleration component. Hence, this acceleration component should not be used as an output Bode response variable, as in the case of $\mathbf{y}_2 = a$ in this example.
2. The remaining components of the ROM-approximated $\hat{\mathbf{x}}$ are correct. Hence, the alternative acceleration output definition, or $\mathbf{y}_1 = \dot{U}$ in this example, gives the correct Bode response result.